

Tristan Dyson

Newcastle, Australia - +61 0435277798 - tristanxdyson@gmail.com
github.com/dargem - linkedin.com/in/tristan-dyson

EDUCATION

UON (University Of Newcastle)

Bachelor of Data Science / Bachelor of Computer Science
Major in Artificial Intelligence
GPA: 7.0 / 7.0

Newcastle, NSW
2025 - Present
Expected 2028

Academic Excellence Scholarship Recognising Outstanding Performance in Undergraduate Studies

Scholarship awarded to top 32 undergraduate students across all programs based on WAM, valued at \$10k

High Performing Students Program - BUSN 1001: Globalisation and International Management

Scheme providing high school students access to university level courses

2022 @ UON

EXPERIENCE

Software Engineer | NUBots Robotics Research Group | C++

nubook.nubots.net

September 2025 - Present

github.com/NUbots

Optimized computer vision algorithms for humanoid soccer robots that compete internationally

- Reduced misdetections in the computer vision pipeline by combining clusters of detections on the same ball.
- Collaborated in a complex 100k+ line codebase alongside a large team using Jira and Git for agile development.

PROJECTS

Benchmark Collection | C++

January 2026 - Present

Benchmarks, verified through statistical testing such as confidence intervals and ANOVAs

github.com/dargem/benchmark_fun

- Designed microbenchmarks for memory layouts (SOA vs AOS), SIMD, branch prediction, and RNG performance
- Showed results were reliable through statistical testing such as confidence intervals and ANOVA with post-hocs
- Achieved up to 10x speedups with SOA + SIMD on uint8_t data and up to ~7.5x gains from parallel execution policies
- Found counterintuitive results where vector<bool> outperform char vectors at scale due to bitpacking reducing L1 cache pressure
- Discovered intricacies like long double SIMD gains disappearing due to lack of AVX2/AVX-512 support for 80-bit floats

Graphics Engine | C++, Java

November 2025 - Present

Procedurally generated ASCII background art using a custom graphics renderer and ECS

github.com/dargem/TUI-Art

- Creating a heavily templated Entity Component System enabling cache-efficient, data-oriented architecture
- Created an efficient graphics rendering pipeline for rasterizing meshes
- Implemented smart diffing engine exploiting viewport movements and line prints to reduce character output by 23%

MMO Combat Simulator | C++, Java

August 2024 - December 2025

Technologies: Springboot, React

github.com/dargem/Vikings-Rise-Simulator

- Deployed a cloud hosted MMO combat simulator with ~200 monthly users
- Derived combat formulas through regression analysis using in-game sampling
- Reconstructed backend combat system with faithful modelling through monte carlo simulations

BabbleFish: NLP Pipeline For Text Translation | Python

June 2025 - November 2025

Technologies: Neo4j, spaCy, networkX, LangGraph, CorEx_Topic

github.com/dargem/BabbleFishv2

- Implemented temporal knowledge graphs for retrieval augmented generation, improving naive semantic searches
- Designed a topic classifier combining spaCy for preprocessing, CorEx_Topic for topic modelling and Gemini for post processing
- Utilized LangGraph for stateful graph based orchestration that allowed the use of complex feedback loops

AWARDS & SKILLS

Placed on the **UON College Commendation list** for 2025

Languages: C++, Java, Python, R, SQL

Tools / Frameworks: Git, Jira, spaCy, LangGraph, Pandas, Eigen, Neo4j

Extracurriculars: Competitive Programming, competing in SPAR (South Pacific Algorithmic Rounds) in practice for ICPC

Interests: Modern C++, operating systems, performant code, computer graphics and data-orientated design